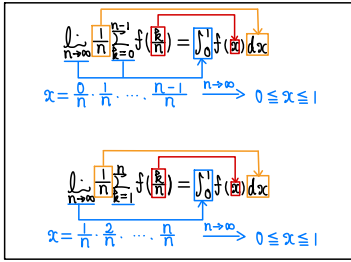


区分的積分と極限①



(例1) 次の極限値を求めよ。

$$\begin{aligned}
 (1) \quad \lim_{n \rightarrow \infty} \frac{1}{n} (\sin \frac{\pi}{n} + \sin \frac{2\pi}{n} + \sin \frac{3\pi}{n} + \dots + \sin \frac{n\pi}{n}) \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \sin(\pi \cdot \frac{k}{n}) \\
 &= \int_0^1 \sin \pi x \, dx \\
 &= \left[-\frac{1}{\pi} \cos \pi x \right]_0^1 \\
 &= \frac{2}{\pi} \quad \text{,,}
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad \lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} \right) \\
 &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n+k} \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{1+\frac{k}{n}} \\
 &= \int_0^1 \frac{1}{1+x} \, dx \\
 &= [\log |1+x|]_0^1 \\
 &= \log 2 \quad \text{,,}
 \end{aligned}$$

$$\begin{aligned}
 (3) \quad \lim_{n \rightarrow \infty} \frac{1}{n^2} \{ (\sqrt{1} + \sqrt{n})^2 + (\sqrt{2} + \sqrt{n})^2 + (\sqrt{3} + \sqrt{n})^2 + \dots + (2\sqrt{n})^2 \} \\
 &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n^2} (\sqrt{k} + \sqrt{n})^2 \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\sqrt{\frac{k}{n}} + 1 \right)^2 \\
 &= \int_0^1 (\sqrt{x} + 1)^2 \, dx \\
 &= \int_0^1 (x + 2\sqrt{x} + 1) \, dx \\
 &= \left[\frac{1}{2} x^2 + \frac{4}{3} x^{\frac{3}{2}} + x \right]_0^1 \\
 &= \frac{17}{6} \quad \text{,,}
 \end{aligned}$$

(例2) 次の極限値を求めよ。

$$\begin{aligned}
 (1) \quad \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \frac{n}{3n^2 + k^2} &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \frac{1}{3 + (\frac{k}{n})^2} \\
 &= \int_0^1 \frac{1}{3+x^2} \, dx
 \end{aligned}$$

ここで、 $x = \sqrt{3} \tan \theta$ とおく

$$dx = \frac{\sqrt{3}}{\cos^2 \theta} d\theta$$

$$\begin{array}{c|c} x & 0 \rightarrow 1 \\ \hline \theta & 0 \rightarrow \frac{\pi}{6} \end{array}$$

であるから

$$\begin{aligned}
 (\text{与式}) &= \int_0^{\frac{\pi}{6}} \frac{1}{3(1+\tan^2 \theta)} \cdot \frac{\sqrt{3}}{\cos^2 \theta} d\theta \\
 &= \int_0^{\frac{\pi}{6}} \frac{\cos^2 \theta}{3} \cdot \frac{\sqrt{3}}{\cos^2 \theta} d\theta \quad \leftarrow 1 + \tan^2 \theta = \frac{1}{\cos^2 \theta} \\
 &= \int_0^{\frac{\pi}{6}} \frac{\sqrt{3}}{3} d\theta \\
 &= \left[\frac{\sqrt{3}}{3} \theta \right]_0^{\frac{\pi}{6}} \\
 &= \frac{\sqrt{3}}{18} \pi \quad \text{,,}
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad \lim_{n \rightarrow \infty} \frac{(\sqrt{n+1} + \sqrt{n+2} + \dots + \sqrt{2n})^4}{1^5 + 2^5 + \dots + n^5} \\
 &= \lim_{n \rightarrow \infty} \frac{\left(\sum_{k=1}^n \sqrt{n+k} \right)^4}{\sum_{k=1}^n k^5} \\
 &= \lim_{n \rightarrow \infty} \frac{n^2 \left(\sum_{k=1}^n \sqrt{1+\frac{k}{n}} \right)^4}{n^5 \sum_{k=1}^n \left(\frac{k}{n} \right)^5} \\
 &= \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{n} \sum_{k=1}^n \sqrt{1+\frac{k}{n}} \right)^4}{\frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n} \right)^5} \\
 &= \frac{\left(\int_0^1 \sqrt{1+x} \, dx \right)^4}{\int_0^1 x^5 \, dx} \\
 &= \frac{\left(\left[\frac{2}{3} (1+x)^{\frac{3}{2}} \right]_0^1 \right)^4}{\left[\frac{1}{6} x^6 \right]_0^1} \\
 &= \frac{32(2\sqrt{2}-1)^4}{2^7} \quad \text{,,}
 \end{aligned}$$